SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

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Course Code:	CSA1407	**Course Title:**	COMPILER DESIGN
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	40% of CIA		
CO1 Statement:	Apply lexical analysis for token specification and recognition using LEX tool. (BL3)		
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Section / Batch:	2023	**Date:**	14-07-2026

Code of Conduct

I M. Vishnu Prasad (192311217) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Q. No	Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)	Total Score (out of 100)
1	4 / 4	4 / 4	4 / 4	4 / 4	3 / 4	96.25
2	4 / 4	3 / 4	3 / 4	4 / 4	4 / 4	88.75
3	3 / 4	4 / 4	4 / 4	4 / 4	3 / 4	91.25
4	4 / 4	4 / 4	3 / 4	3 / 4	3 / 4	85
5	3 / 4	3 / 4	4 / 4	4 / 4	4 / 4	90
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	90.25
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

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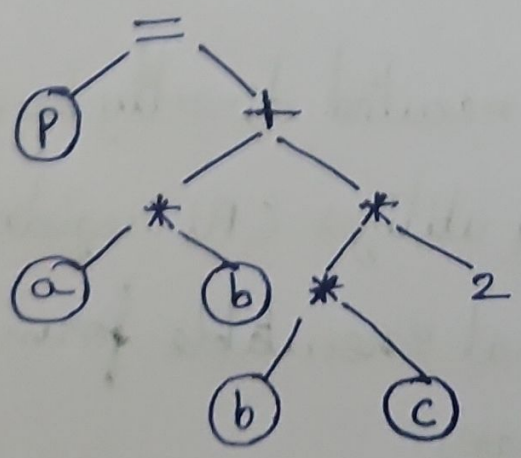
a) lexical analysis :- $P = (a*b)+(b*c)*2$

Lexeme	Token type
P	Identifier
=	Assignmentoperator
(	left parenthesis
a	Identifier
*	Multiplication operator
b	Identifier
)	parenthesis
+	Addition operator
(	Parenthesis
b	Identifier
*	Mul. operator
c	identifier
)	paretheris
*	Mul. operator
2	Integer

b) Syntax tree :-



c) Intermediate code generation :-

$t_1 = a * b$

$t_2 = b * c$

$t_3 = t_2 * 2$

$t_4 = t_1 + t_3$

$p = t_4$

d) Code optimization :-

$t_1 = a * b$

$t_2 = b * c$

$t_3 = t_2 + t_2$

$p = t_1 + t_3$

e) Final target code :-

```
MOV R1, a
MUL R1, b
MOV R2, b
MUL R2, C
SHL R2, 1
ADD R1, R2
MOV P, R1
```

Significance :-

→ It is the machine dependent code produced by the compiler.

→ It can be executed directly by the processor.

→ It efficiently utilizes CPU registers & instructions.

→ It is the final executable form generated after all compiler phases.

a) $(\varepsilon + aa^*)^*$

All strings consisting of zero or more a's

$$L = \{a^n \mid n \geq 0\}$$

- ε
- a
- aa
- aaa
- aaaa

b) $(a^* + b^*)abb$

Any combination of a's & b's followed by abb.

$$L = \{wabb \mid w \in \{a,b\}^*\}$$

- abb
- aabb
- babb
- aaabb
- bbabb

c) $(a^*b^*)^* aba (a^* + b^*)^*$

All strings over {a,b} containing the substring "aba".

- aba
- aaba
- abab
- abaa
- babab

d) $(a+ab)^*$

strings formed by repeating either a or ab any number of.

- ε
- a
- ab
- aa
- aab

e) $(aba)^* + (a^* b^*)^*$

All possible strings over {a, b}, including the empty string.

- ε
- a
- b
- ab
- ba

Q3) i)

S.No	Tokens
1	main
2	(
3	)
4	{
5	int
6	a
7	=
8	10
9	;

10	char
11	b
12	=
13	"abc"
14	;
15	int
16	c
17	=
18	30
19	;
20	char
21	d
22	=
23	"xyz"
24	;
25	int
26	m
27	=
28	40.5
29	;
30	}

∴ Total number of tokens = 30//

iii) lexical errors :-

$x = 1xab;$

Error = 1xab

∴ Invalid numeric constant. Identifiers cannot start with a digit and 1xab is neither a valid integer nor a valid hexa decimal constant.

Q4) i) Languages of all strings having an even number of 0's

$$R.E = 1^*(01^*01^*)^*$$

ii) Language of all strings containing at least one 1

$$R.E = (0+1)^*1(0+1)^*$$

iii) Language of all strings in which both the number of 0's & the number of 1's are odd

$$R.E = (00+11)^*(01+10)(00+11)^*((01+10)(00+11)^*(01+10)(00+11)^*)^*$$

iv) Language of all strings that start with 1 & end with 0.

$$R.E = 1(0+1)^*0$$

v) Language of all strings that do not contain consecutive 1's.

$$R.E = (0+10)^*(\varepsilon+1)$$

S.No	Token
1	int
2	var 123
3	=
4	12345
5	;
6	float
7	var 456
8	=
9	123.45
10	;
11	if
12	(
13	var123
14	==
15	var 456
16	)
17	{
18	return
19	;
20	}

∴ Total number of tokens = 20//

b) Keywords :-

→ int
→ float
→ if
→ return

Operators :-

→ =
→ =
→ = =

Separators :-

→ ;
→ ;
→ (
→)
→ {
→ ;
→ }

Identifiers :-

→ Var123
→ Var456
→ Var123
→ Var456

Integer literals :-

→ 12345

Floating point literals :-

→ 123.45

c)

Token	Transition
int	3
var123	6
=	1
12345	5
;	1
float	5
var456	6
=	1
123.45	6

Token	Transition
123.45	6
;	1
if	2
(	1
var123	6
==	2
var456	6
)	1
{	1
return	6

Token	Transition
;	1
}	1
Total	70

∴ FSM transitions = 70